

ADAPT

Survival-Analytic Command-Channel Prognostics for Degenerative-Condition Accessibility Interfaces

Proportional-Hazards Forecasting of Command-Class Failure, Triggering Proactive (Pre-Failure) Accessibility Vocabulary Migration

Hardware: Muscle BioAmp Shield v0.3 (1 ch) | Arduino | ESP32 | SG90 Servo — ALL OWNED

DOMAIN:	Assistive Technology — Adaptive HCI / Prognostic Interfaces
CLAIM SCOPE:	NARROWED post-search — see Section 3 for what was cut and why
VERIFIED AGAINST:	US11029758B2 (dynamic remapping); US10777200 (cognitive adaptation)
CORE METHOD:	Cox proportional-hazards model on per-command reliability covariates
HARDWARE CHANGE:	NONE — same SIGNAL hardware; this is a software/algorithm layer above it
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1. Abstract

This document discloses a narrowly-scoped software architecture layer, ADAPT, that sits above a single-channel EMG accessibility interface (the previously specified SIGNAL system) and forecasts, using a proportional-hazards survival model, when an individual command class within a burst-pattern vocabulary will become unreliable for a specific user, due to progressive neuromuscular decline. On crossing a calibrated hazard threshold, ADAPT proactively migrates the affected command to an alternative encoding before the user experiences functional failure, rather than waiting for failure to occur and reacting after the fact.

This claim was deliberately narrowed after a documented prior art search (Section 3) revealed that general dynamic EMG/BCI command remapping is already patented (US11029758B2, granted 2021, active to 2039), and that ability-aware adaptive interaction for impaired users is an active patent category more broadly (e.g. US10777200). What remains unclaimed, as far as this search could determine, is the specific combination of: (a) a formal time-to-failure hazard model applied to individual EMG command-class reliability, (b) calibrated specifically against documented physiological decline patterns in progressive neuromuscular conditions, and (c) used to trigger migration before failure rather than in reaction to detected context change or already-degraded performance.

CAUTION: *This document does not claim novelty over general adaptive interfaces, general EMG remapping, or general predictive maintenance / survival analysis as mathematical techniques — all of those are established prior art and are cited as such throughout. The claim is the specific combination and its application to this specific problem.*

2. Origin Note — How This Claim Was Narrowed

This section exists because the claim in this document is substantially smaller than the ADAPT concept as originally proposed, and that difference matters for understanding what is and is not defensible.

2.1 What Was Originally Proposed

The original ADAPT concept proposed four layers: (1) intent extraction from EMG bursts, (2) a capability profiler tracking contraction duration, repeatability, fatigue, and timing variance into a single score, (3) an adaptive command mapper that automatically restructures the command vocabulary as specific commands become unreliable, and (4) a "Universal Accessibility Bus" routing abstract intents (SELECT/BACK/HOME) to multiple device types (AAC, wheelchair, smart home) through device-specific adapters. A further addition proposed an "Accessibility Digital Twin" forecasting which command would fail next, weeks in advance.

2.2 What a Documented Search Found

A two-part prior art search (full record in Section 3) found that the Universal Accessibility Bus concept (layer 4) and the core remapping mechanism (layer 3, as a general mechanism) are both already covered by granted, active patents. Specifically, US11029758B2 already claims detecting a change in computing-device operation context and remapping a muscle-activity-to-command mapping in response, including explicit coverage of EMG-based control, wheelchair control, and multiple downstream applications through a single command vocabulary. This is functionally what layers 3 and 4 proposed, already granted and active until 2039.

2.3 What Survives

The one element not covered by '758 or by the adjacent cognitive-adaptation patent found in search: '758's remapping trigger is external operation context (which application is running, an incoming message, a location change, user history). It is not triggered by a forecast of the input channel's own physiological reliability decaying over time. That is the narrow seam this document now occupies, and it is the only thing claimed going forward. The Capability Profiler concept survives, but reframed rigorously as a formal hazard model rather than a single ad hoc "score," because an unspecified scoring function is not a defensible claim element on its own.

3. Prior Art Search — Full Documented Record

Two searches were conducted. Both are reproduced here in full so the scope decision in Section 4 can be independently checked rather than taken on faith.

3.1 Search One — Adaptive/Degeneration-Aware EMG Remapping

Reference	What It Covers	Relevance / Effect on Claim Scope
US11029758B2	Dynamic command remapping for human-computer interface (2021, active to 2039)	Claims defining a muscle-activity-command mapping, detecting a change in computing-device operation context, and remapping the mapping in response, explicitly covering EMG sensors on residual muscle and explicitly covering wheelchair control as a downstream application. DIRECTLY BLOCKS: general dynamic EMG command remapping, and remapping-to-multiple-devices as a general mechanism.
WO2019099831A1 / US11284844B2	EMG assistive communications device, context-sensitive UI	Bioelectrical signal monitored, switch threshold triggers UI events; context-sensitive but context is application/situational, not a forecast of channel reliability decline. Overlaps with general EMG-switch space, not with prognostic remapping specifically.
User adaptation in Myoelectric Man-Machine Interfaces (academic, not a patent)	Regression vs. classification approaches to EMG under non-stationarity	Academic, not a patent; studies user-side adaptation to signal drift, not system-side forecasting of decline. Relevant as technical background, not as a blocking prior art.

3.2 Search Two — Command Vocabulary Simplification for Declining Ability

Reference	What It Covers	Relevance / Effect on Claim Scope
US10777200B2	AI for mitigating effects of long-term cognitive conditions on patient interactions	Recognizes a patient's diminished cognitive capability and translates their personalized communication into a recognizable form for others, and automates device interaction based on that translation. Cognitive, not motor; output-side translation, not input-side command vocabulary forecasting. Confirms the general category (ability-aware adaptive assistive systems) is active and patrolled, raising the bar for how narrow a new claim must be.

US9208661B2 (cognitive ability context system)	Context-dependent application/event activation for people with various cognitive ability levels	Dynamically adjusts speaking speed, vocabulary, and grammatical complexity of system-produced speech based on a stored cognitive-ability data structure, with caregiver input. Again output-side (speech complexity), externally-assessed ability rather than self-measured signal-derived trend, and cognitive rather than motor.
EMG degradation / electrode-shift / classifier-decay literature (multiple academic sources)	Documents that EMG classifier accuracy degrades from electrode shift, fatigue, and non-stationarity, and that this degradation is measurable and partially predictable	Confirms the underlying physiological/signal phenomenon is real and well-studied. All located mitigation strategies (feature alignment, transfer learning, recalibration) aim to preserve the SAME command set despite drift. None forecast individual command-class time-to-failure to trigger vocabulary migration. This is the gap.

3.3 Conclusion of the Search

No patent or publication located in either search applies formal survival/hazard modeling to individual accessibility command-class reliability as a function of a measured physiological decline trend, for the purpose of triggering proactive vocabulary migration ahead of functional failure. This is treated as the working novelty boundary for the remainder of this document. It is a narrow boundary, and the claims in Section 17 are drafted to stay strictly inside it.

4. What Is and Is Not Being Claimed

Element	Status
Forecasting time-to-failure of a specific EMG command class using a hazard model fitted on session-level reliability covariates	CLAIMED — core of this document
Triggering vocabulary migration before the forecast failure point, rather than after failure is observed	CLAIMED — the proactive-vs-reactive distinction is the key differentiator from '758
The specific covariate set used (defined in Section 7) tailored to single-channel EMG burst classification	CLAIMED as an implementation detail supporting claim 1
Remapping a command from one muscle-activity pattern to another in general	NOT CLAIMED — covered by US11029758B2
Routing abstract commands to multiple downstream device types through adapters	NOT CLAIMED — functionally covered by US11029758B2's multi-device claims
Survival analysis / Cox proportional hazards as a mathematical technique	NOT CLAIMED — established statistical method, decades old, used here as applied, not invented
Any specific biological mechanism of motor neuron decline	NOT CLAIMED — relies on published clinical EMG literature, cited not asserted

5. Problem Statement

The SIGNAL system (prior document) gives a switch-access user multiple commands from one EMG site via burst-pattern classification (SHORT, LONG, DOUBLE). This works as a static system. It does not address what happens when the underlying muscle's signal quality is not static, which is the actual situation for the population this is built for.

5.1 Decline Is Not Hypothetical, It Is Documented and Measurable

Clinical EMG literature on motor neuron disease is explicit that decreased recruitment of motor units is often an early and sensitive indicator of declining neuromuscular function, observable directly on EMG well before complete functional loss. Separately, the myoelectric-control literature documents that classifier misclassification rates rise measurably with electrode shift and signal non-stationarity, with reported figures in the 15 to 35 percent misclassification range depending on shift magnitude and electrode configuration. Both bodies of evidence point to the same operational fact: command reliability on a single EMG channel is not constant over the timescale relevant to a degenerative condition, and the decline is gradual and trackable rather than sudden.

5.2 What a Static System Does When a Command Degrades

Nothing, until it fails outright. A user attempting LONG (a sustained contraction) finds it works 90 percent of the time one month and 40 percent of the time five months later, with no system response to that decline until the user effectively loses the function. There is no graceful transition; the user experiences increasing frustration and failed attempts before anyone, including a remote caregiver, notices a pattern.

5.3 What a Reactive System (Like '758) Does

US11029758B2 remaps in response to detected operation context changes, but its triggers are external (application state, incoming events, user history) and do not encode the user's own physiological trend over weeks or months. Even if its general remapping mechanism were repurposed for this use case, it would still only react after a degradation is severe enough to register as a context change in the moment, not forecast it in advance.

5.4 The Gap

A system that tracks each command class's reliability over time as a formal time-to-event process, estimates a hazard of failure, and migrates the command to an alternative encoding while the original is still mostly working, rather than after it has stopped working, has not been found in the prior art search. This is the gap ADAPT occupies.

6. Theoretical Grounding — Why Survival Analysis Is the Correct Model

This section exists to ground the central claim in established statistics rather than an invented formula, the mistake made in an earlier draft of a different system in this project. Every equation in this document is a standard, citable statistical method, applied to a new covariate set and a new application, not a new mathematical invention.

6.1 Why Not Simple Trend Extrapolation

A naive approach (linear regression on accuracy-over-time, extrapolated to a threshold crossing) was the original framing and has two real problems. First, it gives a point estimate with no calibrated uncertainty, so the system cannot distinguish a confident forecast from a noisy guess. Second, it does not naturally handle the fact that most observation periods end with the command still working (censored data, in survival-analysis terms) rather than with an observed failure event — naive regression on accuracy does not have a principled way to use that information.

6.2 Why Survival Analysis Fits This Problem Exactly

Time-to-event modeling with censoring is the established statistical framework for exactly this situation: most industrial sensor-degradation and predictive-maintenance literature frames remaining-useful-life estimation as a survival problem precisely because most monitored units are still functioning (censored) at the time of observation, and the goal is to estimate a hazard function describing instantaneous failure risk as a function of accumulated wear or a covariate trend, not a single point-in-time accuracy number.

- **Direct structural analogy:** a command class's accuracy and timing-variance over sessions plays the same role a bearing's vibration spectrum or a battery's discharge curve plays in mechanical RUL estimation — an observable covariate trend correlated with an unobserved underlying degradation process.
- **Censoring is the normal case, not the exception:** on any given day, the overwhelming majority of command classes for the overwhelming majority of users have NOT failed yet. A model that cannot use "still working as of today" as informative data (which plain trend extrapolation cannot do principled-ly) is throwing away most of the available signal.
- **Hazard functions support exactly the threshold-crossing decision ADAPT needs:** rather than asking "what will accuracy be in 3 weeks" (point estimate, brittle), the system asks "what is the instantaneous probability of failure given everything observed so far" (a hazard rate), which is the quantity that should trigger a proactive migration decision under uncertainty.

6.3 The Specific Model: Cox Proportional Hazards

The Cox proportional hazards model is chosen specifically because it does not require assuming a parametric shape for the baseline hazard (unlike a Weibull or exponential model), which matters here because there is no established baseline failure-time distribution for EMG command-class reliability in degenerative conditions — the literature establishes that decline happens and is measurable, not what shape its time-to-failure distribution takes. Cox's model lets covariates (the per-command health indicators defined in Section 7) scale a flexible, non-parametric baseline hazard, which is the right level of assumption given what is and is not established in the literature.

7. The Health Indicator — Defining a Measurable Per-Command Covariate

Before any survival model can be fit, the system needs concrete, computable per-session covariates. These are defined precisely here, not left as a narrative placeholder.

7.1 Per-Command, Per-Session Observables

Covariate	Definition	Why It's Informative
Accuracy(c, t)	Fraction of attempted instances of command c, classified correctly by a supervised confirmation (caregiver-tagged or self-corrected) during session t	Direct functional measure
MeanRMS(c, t)	Mean RMS amplitude of bursts classified as command c during session t	Tracks raw signal strength decline (motor unit dropout)
DurVar(c, t)	Coefficient of variation of burst duration for command c during session t (std-dev / mean of burst length)	Tracks loss of voluntary timing control, especially relevant for the LONG/HOLD command class
InterAttemptGap(c, t)	Median time between repeated attempts of command c within a session (proxy for user-perceived difficulty / fatigue)	Behavioral proxy: users unconsciously slow down or hesitate before commands they're starting to find unreliable
FalseNeg(c, t)	Count of attempts of command c that produced CMD_NONE (no match) per session	Direct functional measure of misclassification, separate from raw accuracy

7.2 The Composite Health Indicator

Rather than combining these into a single ad hoc score (the original, underspecified "Capability Score"), each covariate is kept separate and entered into the hazard model individually, which is standard practice in proportional-hazards modeling and avoids the information loss and arbitrary-weighting problem of collapsing multiple signals into one number before fitting. The model itself learns the relative weighting of each covariate via its fitted coefficients (Section 8), rather than the weighting being hand-specified in advance.

7.3 Observation Cadence and Censoring Definition

A session is defined as one calibration-and-use period (Section 11 details this on the existing SIGNAL hardware). For each command class c, the event of interest is AccuracyFailure(c): the first session in which Accuracy(c, t) falls below a clinically and functionally meaningful floor (default 50 percent, configurable), sustained across two consecutive sessions to avoid triggering on a single noisy session. If this event has not occurred as of the most recent session, command c is right-censored at that session for that user — this is the normal, majority case and is handled natively by the Cox model rather than discarded or imputed.

8. The Forecasting Model — Cox Proportional Hazards, Fully Specified

This is the actual mathematical content of the claim. Every symbol is defined.

8.1 The Hazard Function

$$h(t \mid x(t)) = h_0(t) * \exp(b_1 * x_1(t) + b_2 * x_2(t) + b_3 * x_3(t) + b_4 * x_4(t) + b_5 * x_5(t))$$

Where $h(t \mid x(t))$ is the instantaneous hazard of command class c failing at session t , given its covariate values at that time; $h_0(t)$ is an unspecified, non-parametric baseline hazard (estimated from the population of observed command-class trajectories, not assumed); x_1 through x_5 are the five covariates defined in Section 7 (Accuracy, MeanRMS, DurVar, InterAttemptGap, FalseNeg), each as a function of session t since they are time-varying, not fixed at baseline; and b_1 through b_5 are the fitted coefficients, estimated from data, representing how strongly each covariate scales failure risk.

8.2 Why Time-Varying Covariates, Not Baseline-Only

A standard Cox model often uses only baseline (session-zero) covariate values. Here, covariates are explicitly time-varying because the entire point is to track trend, not a single snapshot. This is an established extension (the time-varying or "extended" Cox model), not a novel statistical contribution — the novelty is in applying it to this covariate set for this purpose, not in the modeling technique itself.

8.3 Parameter Estimation

Coefficients $b_1 \dots b_5$ are estimated by maximizing the Cox partial likelihood, the standard estimation procedure for proportional hazards models, which conveniently does not require specifying $h_0(t)$ at all during estimation — only the relative ordering of failure times and the covariate values at each, which is exactly the data available from a population of users' command-class histories pooled together for initial model fitting, then refined per-user as more of that individual's session history accumulates.

8.4 From Hazard to an Actionable Forecast

The hazard function gives an instantaneous rate, not directly a date. The cumulative hazard and the corresponding survival function are derived in the standard way:

$$H(t) = \text{integral from } 0 \text{ to } t \text{ of } h(u) \, du$$

$$S(t) = \exp(-H(t))$$

$S(t)$ is the estimated probability that command class c has NOT yet failed by session t . The system computes a forecast failure window as the session at which $S(t)$ first drops below a configurable confidence threshold (default 0.7, meaning a 30 percent estimated cumulative failure probability), rather than waiting for a point estimate alone — this ties the actionable trigger directly to a calibrated probability rather than an arbitrary score crossing an arbitrary line, which was the unspecified part of the original pitch.

9. Worked Numerical Example

To make Section 8 concrete rather than purely symbolic, here is a illustrative worked trajectory for a single hypothetical command class (LONG/HOLD) across six monthly sessions. These are illustrative numbers consistent with the ranges reported in the EMG-decline literature cited in Section 5.1, not measured data from an actual deployment (no deployment has occurred yet — this is a worked example for documentation purposes).

Session t	Accuracy	MeanRMS	DurVar	InterAttemptGap (s)	S(t) estimate
1	0.94	210	0.08	1.2	0.96
2	0.91	195	0.11	1.4	0.93
3	0.85	170	0.16	1.9	0.86
4	0.74	140	0.24	2.6	0.71
5	0.61	110	0.33	3.8	0.49
6	0.46	90	0.41	5.1	0.27

Reading this trajectory: accuracy crosses the 50 percent floor between session 5 and session 6. Under the proactive policy (Section 10) with a default S(t) trigger of 0.7, the system would have flagged LONG for migration as of session 4 (where the illustrative S(t) of 0.71 first approaches the threshold), roughly one full session-cycle (here, one month) before the command actually crossed the functional failure floor at session 6. That lead time, not the exact numbers, is the operational claim: migrate while the command is still mostly usable, not after it has already failed the user in practice.

10. The Migration Policy — From Hazard Estimate to Action

A forecast alone does nothing; this section specifies exactly what the system does once $S(t)$ for a command crosses the action threshold.

10.1 Substitution Priority Order

Not all commands are equally important to preserve. A fixed priority order, configurable per deployment but defaulting to clinical common sense, governs which functions survive longest when multiple commands are flagged:

Priority	Command	Rationale
1 (highest)	SHORT (Select / affirmative)	The single most load-bearing command; every other command can in principle be re-derived from repeated SHORTs if absolutely necessary, so this is protected last
2	DOUBLE (Back / negative)	Second most load-bearing; binary yes/no coverage is preserved as long as possible
3 (lowest)	LONG / HOLD (Home / tertiary)	Requires sustained voluntary contraction, which the EMG-decline literature identifies as typically the first capability lost; migrated earliest and most readily

10.2 Substitution Rule

When a lower-priority command is flagged, it is replaced by a compound pattern built from higher-priority, currently-healthy commands, exactly as illustrated narratively in the original ADAPT pitch (LONG migrating to SHORT-SHORT, then further compounding as needed). Formally: if command c_k is flagged and c_j ($j < k$, higher priority) has the highest current $S(t)$ among healthy commands, c_k is remapped to a fixed repetition pattern of c_j (e.g., a double-tap of c_j), and the system enters a brief guided practice mode (Section 11) so the user's first few attempts at the new pattern are met with a wider tolerance window before tightening to normal classification parameters.

10.3 What Happens If the Highest-Priority Command Itself Is Flagged

CAUTION: If *SHORT* itself, the highest-priority and structurally simplest command, begins showing a rising hazard, this indicates the single EMG channel as a whole is approaching the limit of what it can support, not just one command class within it. ADAPT's correct response in this case is NOT to attempt further vocabulary compression on a single channel, but to flag this explicitly to the user and any configured caregiver as a signal that an alternative input modality (different muscle site, different sensing approach entirely) should be evaluated. This is stated explicitly because silently degrading the system's only remaining reliable command would be unsafe; the honest design response to a fundamental floor is to surface it, not to engineer around it indefinitely.

11. System Architecture (Same Hardware as SIGNAL)

No hardware changes from the previously specified SIGNAL build. ADAPT is purely a software layer that consumes SIGNAL's existing telemetry stream and adds a model-fitting and decision layer above it.

Layer 1 — Acquisition (unchanged)	Muscle BioAmp Shield onboard AFE -> A0. Same as SIGNAL.
Layer 2 — Burst Decode (unchanged)	Arduino: DC-block, RMS, TKEO, burst-pattern FSM. Same as SIGNAL.
Layer 3 — Session Logger (NEW, lightweight)	Per-attempt outcome logged with covariates (Section 7) to local storage each session.
Layer 4 — Hazard Model (NEW, host-side)	Cox PH model fit/updated periodically (not real-time); outputs S(t) per command class.
Layer 5 — Migration Engine (NEW)	Applies Section 10 policy; updates the firmware's active command-to-class mapping.

Layers 1-2 are exactly SIGNAL's existing firmware, unmodified. Layers 3-5 are new and live almost entirely on the host/ESP32 side, not on the Arduino, because Cox model fitting is not something an Arduino-class microcontroller should attempt — this is explicitly NOT real-time and does not belong in the 1kHz control loop.

12. Data Pipeline & Where Computation Runs

```

[Arduino – existing SIGNAL firmware]
[Per-attempt: command, success/fail, RMS, duration]
    | Serial, decimated
[ESP32 or host PC – Session Logger]
[Appends to local session log, one row per attempt]
    |
    (end of session OR nightly batch)
    |
[Host PC – Python: covariate aggregation]
[Computes Accuracy/MeanRMS/DurVar/Gap/FalseNeg per command, per session]
    |
[Host PC – Python: Cox PH model fit/update]
[lifelines library; outputs S(t) per command class]
    |
    IF S(t) < threshold for any command:
    |
[Migration Engine]
[Selects substitute pattern per Section 10 policy]
[Pushes new mapping table back to Arduino over serial]
    |
[Arduino – burst classifier now uses updated mapping]
[Guided practice mode active for N sessions on the new pattern]

```

The critical design point: the Arduino's real-time control loop (burst detection, servo actuation) is never blocked on or modified by model fitting. The Arduino only ever receives a simple updated lookup table (which burst pattern maps to which logical command); it has no awareness of hazards, survival curves, or covariates. This keeps the safety-critical, low-latency path exactly as simple and auditable as SIGNAL's original design, with all statistical complexity confined to the non-real-time host layer.

13. Software — Model Fitting & Inference (Python)

Built on the lifelines library, a well-established, peer-reviewed-adjacent open source survival analysis package implementing the Cox proportional hazards model exactly as specified in Section 8 — this is explicitly using an existing, correct implementation of an established statistic, not a hand-rolled approximation.

```
# requirements: lifelines pandas numpy
import pandas as pd
from lifelines import CoxTimeVaryingFitter

# session_log columns (one row per command, per session, per user):
#   user_id, command, session_t, accuracy, mean_rms, dur_var,
#   inter_attempt_gap, false_neg_count, event_failure, start, stop
#
# 'event_failure' = 1 if accuracy < FLOOR for 2 consecutive sessions
#                   ending at this row, else 0 (right-censored)
# 'start'/'stop' = session interval bounds, required by the
#                 time-varying Cox fitter's data format

def fit_hazard_model(session_log: pd.DataFrame, command: str):
    df = session_log[session_log['command'] == command].copy()
    ctv = CoxTimeVaryingFitter(penalizer=0.1) # small L2 penalty
                                              # for stability with
                                              # limited per-user data

    ctv.fit(
        df,
        id_col='user_id',
        event_col='event_failure',
        start_col='start',
        stop_col='stop',
        # covariates, matching Section 7 exactly:
        # accuracy, mean_rms, dur_var, inter_attempt_gap, false_neg_count
    )
    return ctv

def forecast_survival(ctv, current_covariates: dict, horizon_sessions=6):
    """
    current_covariates: dict with the 5 covariate values as of the
    most recent session for this user/command.
    Returns S(t) for t = 1..horizon_sessions ahead.
    """
    import pandas as pd
    future = pd.DataFrame([current_covariates] * horizon_sessions)
    future['start'] = range(horizon_sessions)
    future['stop'] = range(1, horizon_sessions + 1)
    surv = ctv.predict_survival_function(future)
    return surv # S(t) curve, one column per forecasted session

def check_migration_trigger(surv_curve, threshold=0.7):
    """
    Returns the first forecasted session index where S(t) < threshold,
    or None if no crossing within the forecast horizon.
    """
    crossed = surv_curve[surv_curve.iloc[:, -1] < threshold]
    if len(crossed) > 0:
        return crossed.index[0]
    return None

# migration_engine.py -- Section 10 policy, implemented
PRIORITY = ['SHORT', 'DOUBLE', 'LONG'] # highest to lowest priority

def select_substitute(flagged_command: str, healthy_commands: list,
                     survival_scores: dict) -> str:
    """
    healthy_commands: commands NOT currently flagged
    survival_scores: {command: current S(t)} for healthy commands
    Returns a compound pattern string, e.g. 'SHORT_SHORT' for LONG.
```



```
"""
flagged_rank = PRIORITY.index(flagged_command)
candidates = [c for c in PRIORITY[:flagged_rank] if c in healthy_commands]
if not candidates:
    # SHORT itself flagged with nothing higher-priority available
    return 'ESCALATE_TO_CAREGIVER' # Section 10.3
best = max(candidates, key=lambda c: survival_scores.get(c, 0))
return f'{best}_{best}' # double-tap pattern

def push_new_mapping(serial_conn, flagged_command, substitute_pattern):
    msg = f'REMAP,{flagged_command},{substitute_pattern}\n'
    serial_conn.write(msg.encode())
    # Arduino firmware (Section 14) parses this and updates its
    # lookup table; enters guided-practice mode for N sessions.
```

14. On-Device Firmware Changes (Arduino, minimal)

The existing SIGNAL firmware (full listing in the prior document) needs exactly two additions: a remappable lookup table instead of a hard-coded CLASSIFY block, and a serial command parser for the REMAP message. Everything else — TKEO, RMS, the FSM states themselves — is unchanged.

```
// addition to SIGNAL firmware: remappable command table
enum LogicalCmd : uint8_t { LCMD_SELECT, LCMD_BACK, LCMD_HOME, LCMD_NONE };

// maps a detected burst pattern -> a logical command
// default mapping == SIGNAL's original SHORT/LONG/DOUBLE behaviour
struct PatternMap {
    Cmd pattern;      // CMD_SHORT, CMD_LONG, CMD_DOUBLE (from SIGNAL)
    LogicalCmd logical;
} activeMap[3] = {
    { CMD_SHORT,  LCMD_SELECT },
    { CMD_LONG,   LCMD_HOME   },
    { CMD_DOUBLE, LCMD_BACK   },
};

// guided-practice state: widens tolerance on a newly-migrated pattern
bool practiceMode = false;
uint8_t practiceSessionsLeft = 0;

// addition: parse REMAP messages from host
void parseRemapCommand(String line){
    // format: REMAP,,
    // e.g.    REMAP,HOME,SHORT_SHORT
    int c1 = line.indexOf(',');
    int c2 = line.indexOf(',', c1+1);
    String flagged = line.substring(c1+1, c2);
    String newPat  = line.substring(c2+1);
    LogicalCmd flaggedLogical = parseLogical(flagged);
    for(int i=0;i<3;i++){
        if(activeMap[i].logical == flaggedLogical){
            // newPat encodes a compound pattern; for the DOUBLE-derived
            // substitute, we reuse CMD_DOUBLE's detection but relabel it
            activeMap[i].pattern = parseCompoundPattern(newPat);
        }
    }
    practiceMode = true; practiceSessionsLeft = 5; // widen tolerance
}

// existing CLASSIFY state now looks up activeMap instead of a
// hard-coded if/else chain -- this is the only control-flow change
LogicalCmd classifyToLogical(Cmd detectedPattern){
    for(int i=0;i<3;i++){
        if(activeMap[i].pattern == detectedPattern) return activeMap[i].logical;
    }
    return LCMD_NONE;
}
```

Practice mode (a widened classification tolerance for a configurable number of sessions after migration) exists so the user's first attempts at a newly-compound pattern are not penalized at the same strictness as a pattern they have used for months — this is a deliberate usability safeguard, not part of the statistical claim itself.

15. Validation Plan

This is the honest part: nothing in this document has been validated on real longitudinal user data, because no such data exists yet. The validation plan below is what would need to happen before any clinical or even confident maker-level claim could be made, and is presented as future work, not as completed evidence.

Question	Method	Status
Model calibration	Requires longitudinal session data from real users over months, ideally including at least some users with a documented progressive condition	Not yet available — primary blocker to validating anything in Section 8-9
Lead-time benefit	Compare proactive migration (ADAPT) against a reactive baseline (migrate only after functional failure) in simulation first, using synthetic decline trajectories grounded in the literature ranges from Section 5.1	Feasible immediately as a simulation study, independent of real user data
False-trigger rate	How often does the hazard model flag a command that does not actually go on to fail (a false positive migration, which has its own usability cost)	Requires either real data or careful synthetic stress-testing
Migration usability	Does guided-practice mode actually result in acceptable accuracy on the new compound pattern within the configured practice window	User study required, even with able-bodied test subjects in early stages

16. Risks, Limitations & Clinical Caveats

Unvalidated model: Every coefficient and threshold in Sections 8-9 is illustrative pending real data; this document should not be read as claiming a working, calibrated system, only a specified, buildable one.

Cold-start problem: A Cox model needs a meaningful number of prior failure events across users to fit reliable population-level coefficients before it can be useful for any single new user; early deployments would need to rely on literature-informed priors, explicitly flagged as low-confidence, until enough data accumulates.

Single-channel ceiling: As stated in Section 10.3, this system manages decline within what a single EMG channel can support; it cannot manufacture capability that is not there, and the correct response when even SHORT is failing is escalation, not further compression.

Not a diagnostic tool: Hazard estimates here describe command-classification reliability, not a clinical assessment of disease progression, and must never be presented to a user or caregiver as the latter.

Data sensitivity: Longitudinal EMG performance data from a person with a progressive condition is sensitive health data; any real deployment needs an explicit data-handling and consent design, which is out of scope for this technical document but is not optional in practice.

False reassurance risk: A command sitting just above the migration threshold could still fail unexpectedly between sessions; the system's confidence threshold is a design parameter, not a guarantee, and should be communicated as such.

17. Draft Provisional Claims

Drafted to stay strictly within the narrowed scope established in Sections 3-4. These are draft claims for attorney review, not final language.

Independent Claim 1 — The Method

A computer-implemented method for managing an electromyography-based accessibility command interface, comprising: (a) recording, for each of a plurality of command classes within a burst-pattern command vocabulary, a set of session-level reliability covariates comprising at least classification accuracy, signal amplitude, burst-duration variance, and inter-attempt timing; (b) fitting a proportional-hazards survival model in which said covariates, evaluated as time-varying inputs across a sequence of recorded sessions, scale a non-parametric baseline hazard function representing the instantaneous risk of said command class becoming unreliable for a given user; (c) computing, from said hazard function, a survival probability estimate for each command class at one or more future sessions; (d) responsive to said survival probability estimate falling below a calibrated threshold for a given command class, automatically substituting said command class with a compound pattern derived from one or more other command classes whose survival probability estimate remains above said threshold, prior to an observed functional failure of the original command class.

Dependent Claims

Claim 2: The method of Claim 1, wherein the proportional-hazards model is a Cox model fitted by partial-likelihood maximization across a population of users prior to refinement on session data specific to the given user.

Claim 3: The method of Claim 1, wherein right-censoring is applied to any command class that has not exhibited the failure event as of the most recent recorded session, and such censored observations contribute to the partial-likelihood estimation.

Claim 4: The method of Claim 1, wherein the substitution of step (d) follows a predetermined priority ordering among command classes, such that a command class of lower priority is substituted before a command class of higher priority when both exhibit a qualifying survival probability decline.

Claim 5: The method of Claim 1, further comprising entering a temporary widened-tolerance classification mode for a configurable number of sessions following substitution of a command class, to accommodate user adaptation to the substitute pattern.

Claim 6: The method of Claim 1, wherein, upon the highest-priority command class itself exhibiting a qualifying survival probability decline, the method generates an escalation notification rather than performing a further substitution, the escalation indicating that the underlying physiological signal source may require an alternative sensing modality.

Claim 7: The method of Claim 1, wherein the session-level reliability covariates further comprise a count of unclassified or ambiguous burst attempts per session for the given command class.

18. References

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END OF TECHNICAL DISCLOSURE

Generated 21 June 2026, 05:14 | Narrowed scope following documented prior-art search | No hardware changes from prior SIGNAL specification